

Shubhadeep Mandal

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EDUCATION

International Institute of Information Technology, Hyderabad Telangana, India Master of Technology in Computer Science and Engineering	July 2025 – Present CGPA: 9.12
Indian Institute of Information Technology, Ranchi Ranchi, India Bachelor of Technology (Hons.) in Computer Science and Engineering	Aug 2020 – Jun 2024 CGPA: 9.05

PROJECTS

Multi-Modal AI-SaaS Platform <i>Next.js 13, TypeScript, Prisma, Stripe, PostgreSQL</i>	🔗
- Engineered a commercial SaaS application orchestrating OpenAI and Replicate APIs for text, code, audio, and image generation. - Implemented customer subscription lifecycles, tiered quotas, and secure webhook event processing via Stripe SDK and Clerk. - Designed performant relational database schemas and data pipelines using Prisma ORM over cloud-managed PostgreSQL.	
IndicLLM: Transformer Pretraining & Reasoning SFT <i>PyTorch, FlashAttention-2, RoPE</i>	🔗
- Pretrained 25M-param decoder-only Transformers from scratch on 1B+ tokens in Hindi and Assamese using bare PyTorch primitives, driving a +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU. - Trained 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility tax and achieving 0.505 Bits-Per-Byte compression across high and low-resource languages. - Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, unlocking a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT.	
BoardBuddy: Real-Time Collaborative Canvas <i>Next.js 14, Convex DB, Liveblocks, TypeScript</i>	🔗
- Architected a collaborative workspace supporting real-time multi-user syncing, freehand sketching, and vector shapes. - Synchronized real-time canvas state and user presence across concurrent client sessions using Liveblocks and Convex DB. - Secured organization-scoped workspaces and role permissions with Clerk authentication and a responsive TailwindCSS UI.	
Peer-to-Peer Distributed File Sharing System <i>C++, Sockets, Multithreading, OpenSSL</i>	🔗
- Built a BitTorrent-style P2P system in C++, achieving an 8.5x speedup on 1GB transfers (193s to 22.8s) across 5 parallel peers. - Engineered out-of-order parallel chunk writing via <code>pwrite()</code> and dynamic bitfield tracking for simultaneous partial seeding. - Architected Primary-Backup tracker failover with state synchronization, session flushing to prevent split-brain, and SHA-1 checks.	

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, Rust, SQL, Bash.
AI & LLMs: Transformers (Pretraining & SFT), FlashAttention-2, RoPE, SwiGLU, Prompt Engineering, PyTorch.
Full-Stack & Cloud: Next.js 14, React, Node.js, Convex DB, Prisma ORM, PostgreSQL, Stripe SDK, REST APIs.
Systems & Architecture: WebSockets, POSIX Sockets, Multithreading, Concurrency, Linux/Unix, Git, Docker.

ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.